

Technical Document #624: Computer Vision - Comprehensive Overview

Technical Document #624: Computer Vision - Comprehensive Overview

Optical character recognition (OCR) converts images of text into machine-readable text. It is used in document digitization and automation.

Image classification assigns a label to an image from a fixed set of categories. CNNs like ResNet and EfficientNet achieve superhuman performance on benchmark datasets.

Semantic segmentation classifies each pixel in an image into a category. It is used in autonomous driving and medical image analysis.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Key Takeaways:

- Visual question answering (VQA) answers questions about images.
- Face recognition identifies or verifies faces in images.
- Video understanding analyzes temporal dynamics in video sequences.